

Critical properties of the 2D Ising Model through MCMC

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Abstract

A numerical study of the two-dimensional Ising model on a square lattice with periodic boundary conditions is presented, simulated using a Metropolis MCMC local update algorithm. The autocorrelation analysis allows the critical slowing down to be characterized, yielding $z = 2.18(8)$. Static properties are studied through Finite-Size Scaling: from the intersections of the Binder cumulant curves, $\beta_{pc} = 0.440751(60)$ is obtained, while fits of the critical observables provide $\nu = 0.982(17)$, $\gamma/\nu = 1.742(04)$, $\beta/\nu = 0.124(01)$ and $\alpha = 0.01(02)$. The results are founded to be compatible with the expected values for the universality class of the two-dimensional Ising model.

1 Ising model and physics observables

The Ising model on a two-dimensional square lattice of volume L^2 with periodic boundary conditions (PBC) is considered. The toroidal topology preserves translational invariance, thereby limiting boundary effects [1, 2]. The energy of a spin configuration σ is determined by Hamiltonian [3]:

$$H[\sigma] = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i \quad , \quad (1)$$

with $s_i = \pm 1$. By setting $h = 0$ and adopting $J = 1$ (energy units) and $k_B = 1$, system exhibits invariance under global $\mathbb{Z}_2$ symmetry ($s_i \rightarrow -s_i$). On finite volumes, this invariance enforces $\langle m \rangle = 0$. The ferromagnetic phase transition is rigorously defined by spontaneous symmetry breaking (SSB), which requires the non-commutativity of the thermodynamic and zero-field limit: $\lim_{h \rightarrow 0^\pm} \lim_{L \rightarrow \infty} \langle m \rangle \neq 0$ [1, 3].

The partition function is given by $Z(\beta) = \sum_{\{\sigma\}} e^{-\beta E[\sigma]}$, where $\beta \equiv 1/T$. In the thermodynamic limit, the analytical critical temperature (Onsager solution) is $\beta_c = \frac{1}{2} \ln(1 + \sqrt{2}) \simeq 0.440687$ [2].

1.1 Macroscopic observables and dynamics

Configurations are analyzed through the magnetization and energy densities:

$$m(\sigma) = \frac{1}{L^2} \sum_{i=1}^{L^2} s_i, \quad e(\sigma) = -\frac{1}{L^2} \sum_{\langle ij \rangle} s_i s_j \quad . \quad (2)$$

The Metropolis algorithm generates a Markov chain that asymptotically samples the Gibbs measure [2–4]. While the energy thermalizes rapidly, the magnetization undergoes severe dynamical slowing down (critical slowing down, CSD). Physically, the relaxation time scales as $\tau \sim \xi^z$; at criticality, $\xi \sim L$, implying a divergence $\tau \sim L^z$ [1, 3].

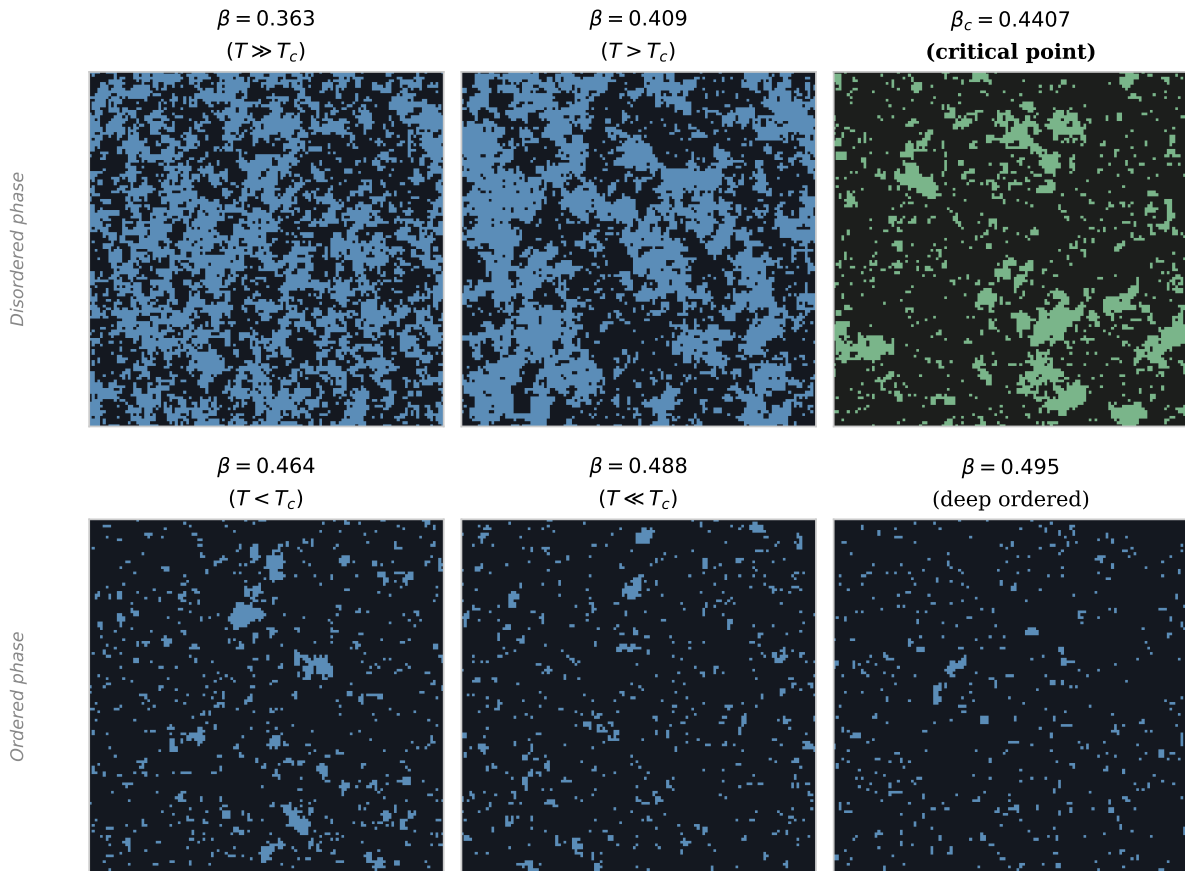


Figure 1: Equilibrium configurations for $L = 128$ at six values of β . The three dominant phases of the model are shown: disordered, critical, and ordered.

Fig. 1 provides a visual illustration of the three phases of the model.

2 Numerical Methodology and Thermalization

Preliminary estimates of the integrated autocorrelation time τ_{int} are essential for calibrating production simulations and for determining the block size in the data blocking procedure [2, 3].

2.1 Sampling of the Parameter Space

A set of lattice sizes $L \in \{10, 16, 24, 32, 48, 64, 96, 128\}$ is adopted. The upper limit $L_{\max} = 128$ represents a compromise to control the computational cost of the Metropolis algorithm, which scales as $\mathcal{O}(L^{d+z})$ [2, 3].

Sampling in β is based on the scaling variable $x = (\beta - \beta_c)L^{1/\nu}$, defining the sampled temperatures as $\beta_i(L) = \beta_c + x_i L^{-1}$ [1, 3].

2.2 Metastability and Tunneling

In the ordered phase ($\beta > \beta_c$), finite volumes allow the dynamics to undergo tunneling transitions between ergodically confined states (Fig. 2).

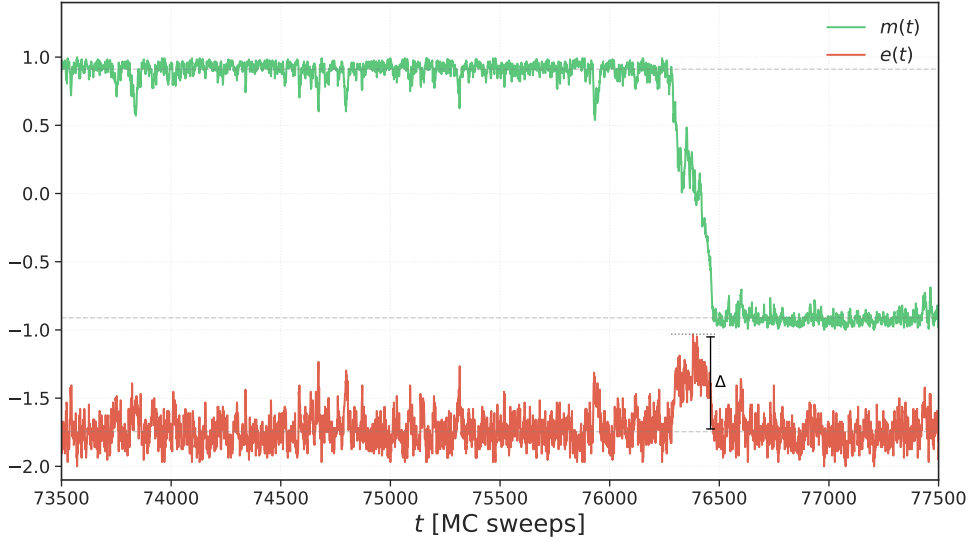


Figure 2: Time evolution of $m(t)$ and $e(t)$ for $L = 16$ at $\beta \simeq 0.50$. The jump in $m(t)$ highlights thermal tunneling across the free-energy barrier Δ . It is observed that the energy increases whenever m approaches zero: configurations with lower energy correspond to larger absolute magnetization. The horizontal lines indicate the asymptotic analytical values.

For increasing volumes, the tunneling probability becomes numerically smaller over finite simulation times [1, 3].

2.3 Verification of Thermodynamic Equilibrium

The attainment of the stationary regime of the Markov chain is tested using three different initial configurations: random ($m \sim 0$), spin up ($m = 1$), and spin down ($m = -1$) (Fig. 3). Statistical compatibility is assessed at the 2σ level between thermalized replicas initialized at $m = 0$ and $m = 1$:

$$|\langle |m| \rangle_{(0)} - \langle |m| \rangle_{(+)}| \leq 2\sqrt{\sigma_{true,0}^2 + \sigma_{true,+}^2} \quad , \quad (3)$$

with the corrected variance $\sigma_{true}^2 = \sigma_{naive}^2(1 + 2\tau_{int,m})$, where $\tau_{int,m}$ is the integrated autocorrelation time of the magnetization [5].

To ensure proper thermalization of the Markov chain and to avoid metastable states, the choice of the number of discarded steps n_{therm} combines the Madras–Sokal (MS) criterion with an empirical inspection of time series. The quantity $n_{therm}^{(0)}$ is defined as the number of steps required for averages computed from opposite initial states ($m = \pm 1$) to converge within a tolerance of 2%. Accordingly, the estimate

$$n_{therm} = \max \left(2n_{therm}^{(0)}, 20\tau_{int,m} \right) \quad . \quad (4)$$

is adopted. The combination of the two methods allows the effective thermalization of the chain to be established unambiguously. A reliable estimate of uncertainties requires

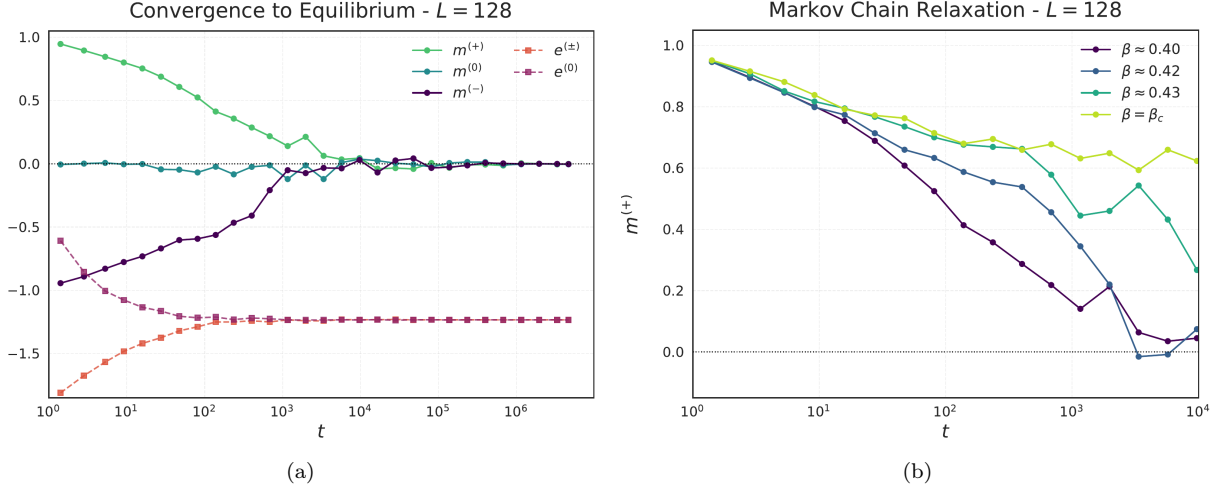


Figure 3: Convergence study of the Markov chain for $L = 128$. (a) Convergence of averages for the three initializations at $\beta \simeq 0.40$. (b) Effect of critical slowing down as $\beta \rightarrow \beta_c$ [4].

a preliminary analysis of $\tau_{int,m}$ [3, 5]. The autocorrelation function for m is defined as [3, 5, 6]:

$$C_m(\Delta t) = \frac{\langle m(t)m(t + \Delta t) \rangle - \langle m \rangle^2}{\langle m^2 \rangle - \langle m \rangle^2}. \quad (5)$$

In the following, the analysis refers to the autocorrelation times and to the corresponding autocorrelation function of the magnetization.

2.4 Madras-Sokal Truncation and CSD

As shown in Fig. 4a, $C(\Delta t)$ decays slowly for $\beta \approx \beta_c$, and as illustrated in Fig. 4b, the decay becomes slower as L increases. To suppress statistical noise in the tails of the discrete integral defining τ_{int} , the sum is truncated at a value W such that $W \geq c \cdot \tau_{int}(W)$ [5], where $c = 6$ (in agreement with the literature [5]) provides a balance between truncation systematic error (Fig. 4c).

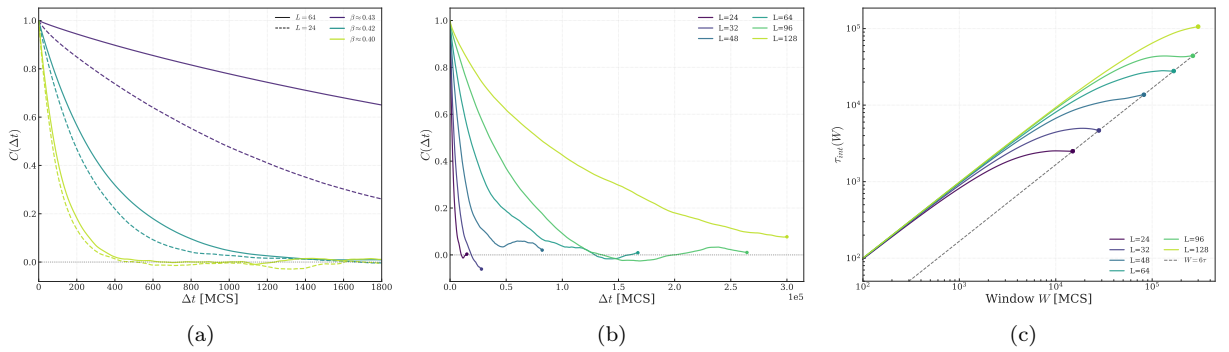


Figure 4: (a) Decay of $C(\Delta t)$ as a function of temperature, highlighting critical slowing down. (b) Autocorrelations at β_c for $L \in$, truncated at W . (c) Saturation of the truncated estimator $\tau_{int}(W)$.

The statistical uncertainty on τ_{int} is computed using the approximation [5, 6]:

$$\sigma_\tau \approx \tau_{int,m} \sqrt{\frac{2(2W + 1)}{N_{meas}}}, \quad (6)$$

which quantifies the error on the estimator τ_{int} as a function of the window W and the total length of the Markov chain.

2.5 Extraction of the Dynamic Exponent z

At criticality, the correlation time scales as $\tau_{int,m}(L) = AL^z$. The results, with errors estimated via the Sokal variance Eq. (6) [5], are reported in Tab. 1.

L	$\tau_{int,m}$	W_{opt}
10	371(46)	2229 16
16	1203(165)	7219 24
24	2497(338)	14984 32
32	4660(630)	27972 48
48	13670(2220)	82027 64
64	25870(5550)	155211 96
96	44000(12300)	263744 128
128	92000(31700)	399999

Tabella 1: Estimates of $\tau_{int,m}$ at β_c , with errors computed using Eq. (6) and optimal window W_{opt} . The values confirm the severity of CSD for the Metropolis algorithm.

A weighted log-log fit (Fig. 5) yields:

$$z = 2.18(8), \quad A = 2.59(68), \quad \chi_{red}^2 = 0.46 \quad . \quad (7)$$

The extracted value is in good agreement with theoretical estimates $z = 2.1667(5)$ for local updates [3]. The value $\chi_{red}^2 = 0.46$ indicates conservative error estimates.

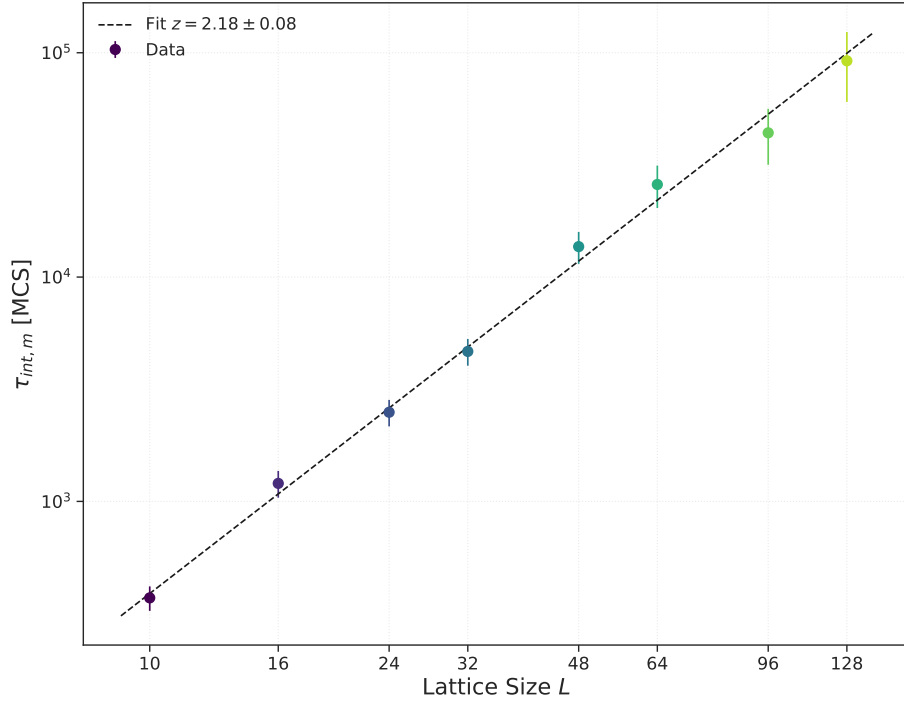


Figure 5: Weighted log-log fit of $\tau_{int,m}$ vs L at β_c . The slope is identified as 2.18(8); error bars reflect MS estimates.

For $L = 96, 128$, CSD reduces $N_{eff} \approx N_{meas}/2\tau_{int}$, deteriorating the signal-to-noise ratio in the tails of $C(\Delta t)$; the MS criterion ensures that the truncation bias remains negligible compared to σ_τ [6].

Validation of Error Estimates: Data Blocking Fluctuations of observables are controlled via Data Blocking: the time series is divided into b blocks, with $b \gg \tau_{\text{int},m}$. The emergence of a plateau in the error σ_b is then assessed. In Fig. 6a, the uncertainty in the estimates of the averages $\langle e \rangle$ and $\langle m \rangle$ for $L = 64$ is shown as a function of k , where k denotes the number of measurements per block b . In Fig. 6b, $\langle m \rangle$ is reported for different values of k .

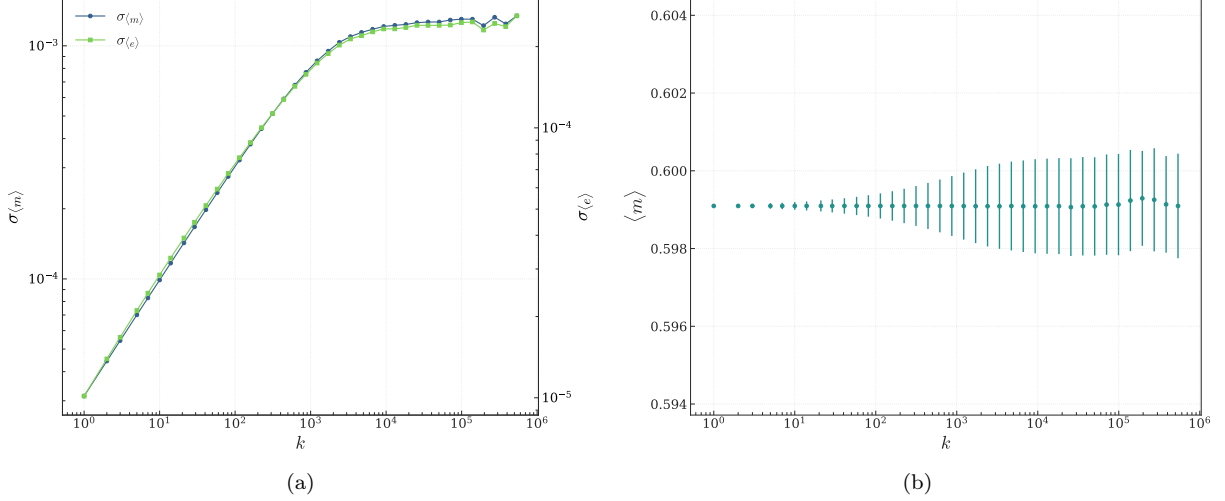


Figure 6: Data blocking for $L = 64$ at β_c [4]. (a) $\sigma_{\langle m \rangle}$ and $\sigma_{\langle e \rangle}$ vs k : saturation for $k \gg \tau_{\text{int}}$. (b) $\langle m \rangle$ vs k .

3 Thermodynamics and Phase Transition

3.1 Probability Distributions and Symmetry Breaking

The ferromagnetic transition clearly emerges from the evolution of the probability density $P(m)$ [1, 3]. In the disordered phase ($\beta < \beta_c$), purely thermal fluctuations generate a unimodal profile centered at zero (Fig. 7a). For $\beta > \beta_c$, the $\mathbb{Z}_2$ invariance and the onset of order shape $P(m)$ into two symmetric peaks at $\pm m_0$ (Fig. 7b).

On finite lattices, stochastic tunneling allows transitions between minima, rigorously enforcing $\langle m \rangle = 0$. However, since the free-energy barrier increases with L , for sufficiently large volumes ($L \geq 64$) such transitions are suppressed, leading to an effective breaking of ergodicity on the time scales of the simulations.

3.2 Magnetic Response: Reduced Susceptibility

Due to $\mathbb{Z}_2$ symmetry, the standard susceptibility $\chi = \beta L^2 \langle m^2 \rangle$, affected by tunneling, is constrained by $\langle m \rangle = 0$. As shown in Fig. 8b, transitions between minima generate in Fig. 8a an artificial saturation $\chi \propto L^2$, which diverges in the thermodynamic limit for any $\beta > \beta_c$. To address this issue, the subtracted susceptibility is adopted [3]:

$$\chi' = L^2 (\langle m^2 \rangle - \langle |m| \rangle^2) \quad . \quad (8)$$

By removing the contribution of the absolute magnetization (which correctly converges to the exact solution, see Fig. 9a), the observable χ' diverges asymptotically only at β_c , developing a peak whose height and width scale with L (Fig. 9b).

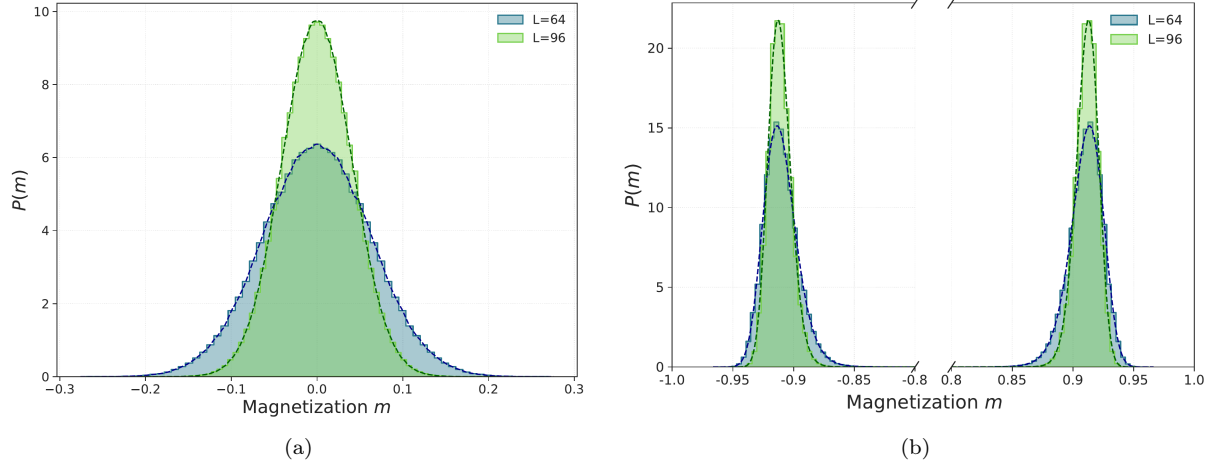


Figure 7: Distributions $P(m)$ for $L = 64, 96$. (a) Unimodal regime at $\beta = 0.35$. (b) Bimodal structure at $\beta = 0.50$.

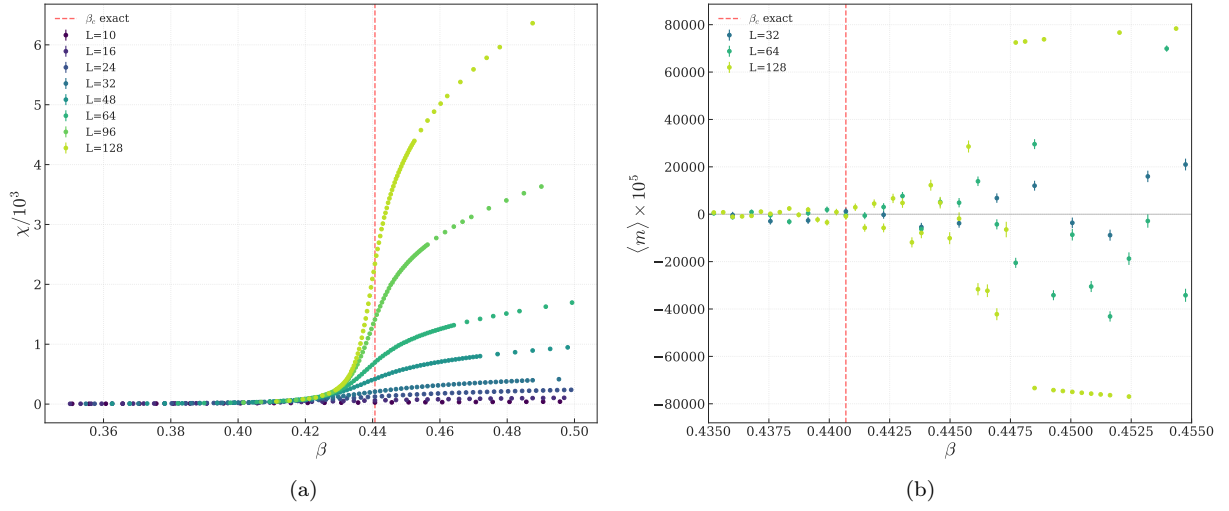


Figure 8: (a) Susceptibility: divergence of $\chi/10^3$ is observed in the ordered phase. (b) Detail of $m(t)$ showing tunneling.

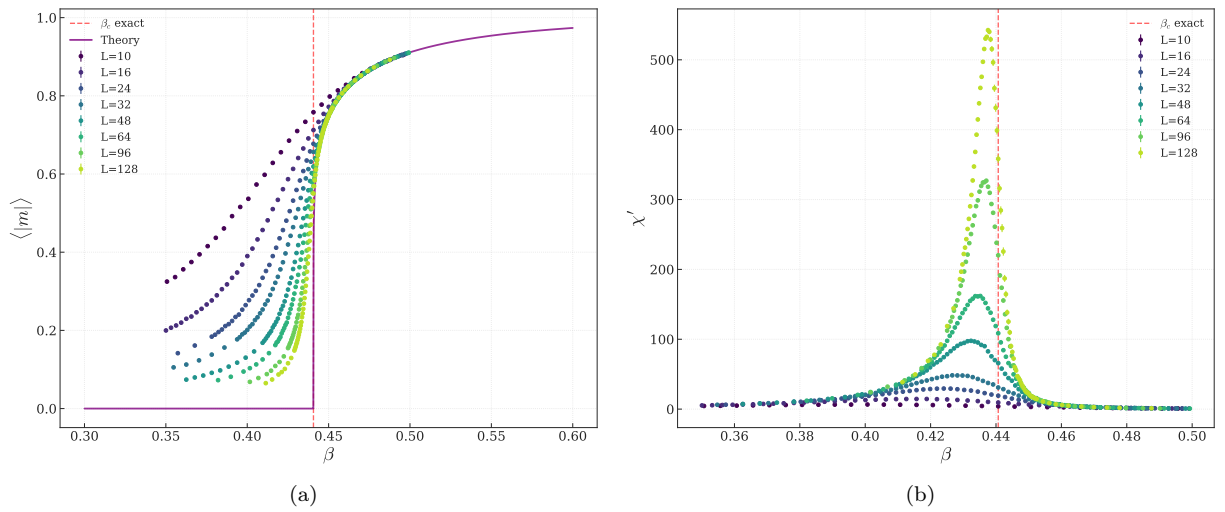


Figure 9: (a) Convergence of $\langle |m| \rangle$ to the analytical solution (purple line). (b) Subtracted susceptibility χ' as a function of β .

3.3 Internal Energy and Specific Heat

The internal energy $\langle e \rangle$ exhibits a smooth inflection point (Fig. 10a), a signature of a second-order phase transition. Accordingly, the specific heat $C_v = L^2(\langle e^2 \rangle - \langle e \rangle^2)$ displays a pseudo-critical peak depending on L (Fig. 10b). The markedly slow growth of $C_{\max}(L)$ qualitatively confirms the expected logarithmic divergence, corresponding to the exponent $\alpha = 0$ for the 2D Ising model [2, 3].

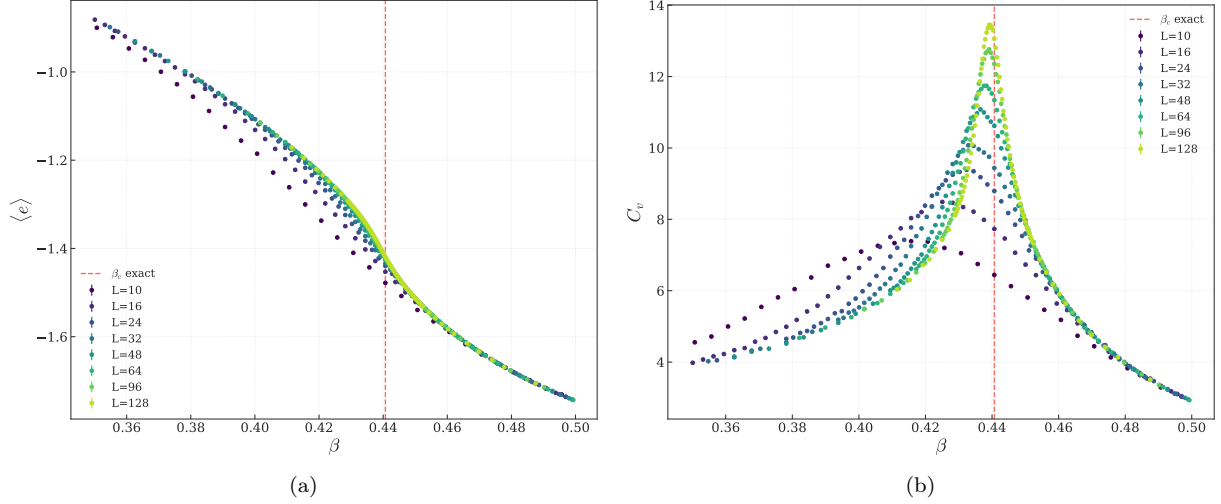


Figure 10: (a) Average energy $\langle e \rangle$ vs β . (b) Development of the specific heat C_v , showing logarithmic divergence for $L \rightarrow \infty$.

3.4 Binder Cumulant and Error Estimation

The Binder cumulant $U_4(\beta, L)$ allows the transition to be located based on its scale invariance [1]:

$$U_4(\beta, L) = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2} . \quad (9)$$

The curves in Fig. 11 satisfy the expected theoretical limits: decay to the Gaussian regime ($U_4 \rightarrow 0$) in the disordered phase and saturation at $2/3$ for $\beta > \beta_c$. The isothermal crossing identifies the pseudo-critical point β_{pc} .

To preserve statistical reliability in the presence of Critical Slowing Down (CSD), uncertainties on primary observables ($\langle m \rangle$, $\langle e \rangle$) are estimated via *Data Blocking* within the decorrelation plateau [2–4]. For derived quantities with nonlinear structure (χ' , C_v , U_4), errors are propagated using the Bootstrap technique [3].

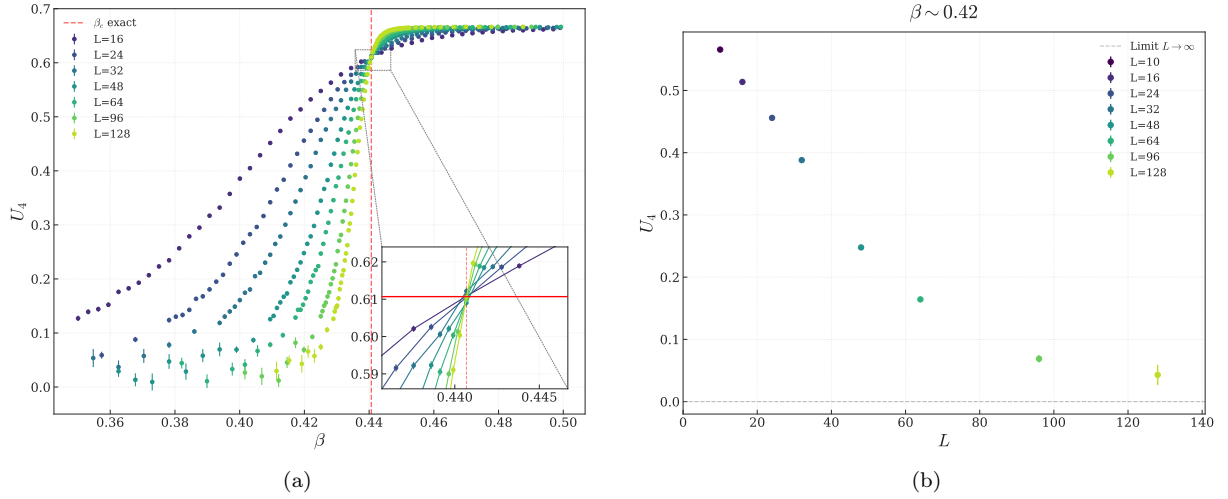


Figure 11: Crossing of the sampled U_4 curves. (b) Decay of U_4 vs L in the disordered phase ($\beta \approx 0.42$).

4 Finite-Size Scaling and Critical Properties

The extraction of the critical exponents was carried out as follows: starting from the pseudocritical point β_{pc} , the exponents were obtained through Finite-Size Scaling (FSS) fits. Subsequently, correlations among the primary exponents were estimated by means of a joint *block-bootstrap* procedure and propagated to the derived quantities.

In a finite volume L^2 with PBC, the singularities of the thermodynamic limit are regularized by the system size L [1]. The standard universal scaling form for an observable $\mathcal{O}$ was adopted [1–3]:

$$\mathcal{O}(\beta, L) = A L^{x/\nu} [1 + B L^{-\omega}] \tilde{f}((\beta - \beta_c) L^{1/\nu}) \quad . \quad (10)$$

The term $B L^{-\omega}$ parametrizes possible sub-leading corrections due to the finiteness of the lattice. Throughout the FSS analysis, a lower cutoff $L_{\min} = 32$ was adopted, so that 5 lattice sizes were used ($L \in \{32, 48, 64, 96, 128\}$). The window used for the critical interpolations is $\beta \in [0.43, 0.45]$. The exact analytical values were used only as an a posteriori reference.

4.1 Pseudocritical Inverse Temperature β_{pc}

The Binder cumulant, $U_4(\beta, L)$, asymptotically intersects the universal value $U_4^* \approx 0.6107$ at the critical point [1, 3]. For each lattice size, a weighted cubic spline was constructed, making it possible to determine the crossing points between pairs of consecutive sizes. Uncertainties were estimated by bootstrap.

Pair (L_1, L_2)	β_{cross}	U^*
(24, 32)	0.440308(692)	0.60933(298)
(32, 48)	0.441082(181)	0.61352(094)
(48, 64)	0.440651(166)	0.60996(141)
(64, 96)	0.440708(084)	0.61061(094)
(96, 128)	0.440759(118)	0.61146(198)

Tabella 2: Crossing points of the Binder cumulant for pairs of consecutive lattice sizes.

The global value β_{pc} was extracted as the weighted mean (with inverse-variance weights) of the crossings in Tab. 2; the final result is:

$$\beta_{pc} = 0.440751(60), \quad U_4^* = 0.6116(6). \quad (11)$$

β_{pc} is compatible with β_c within 1.08σ , and U_4^* is compatible within 1.55σ with the theoretical value. The crossing behavior and the progressive stabilization toward the critical point are shown in Fig. 12.

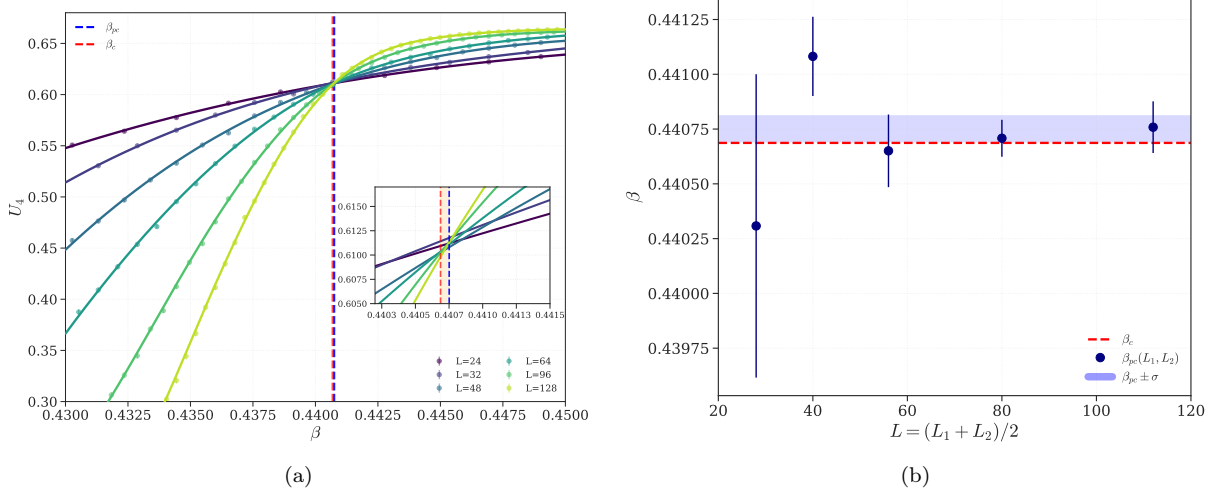


Figure 12: (a) Crossing of the Binder cumulant splines. (b) Convergence of $\beta_{pc}(L_1, L_2)$ to the theoretical limit.

4.2 Exponent ν

The slope of U_4 evaluated at the critical point obeys the scaling $S(L) = |\partial_\beta U_4|_{\beta_{pc}} \sim L^{1/\nu}$. The derivatives were computed analytically from the splines at β_{pc} . The slopes were treated by bootstrap.

An F-test was used to study the hypothesis $B = 0$ for the sub-leading term $B L^{-2}$ in Eq. (10). Fisher's test yielded $p_F = 0.076 > 0.05$, so the test does not allow rejection of the *leading-only* model, which gave:

$$\frac{1}{\nu} = 1.019(17) \quad \implies \quad \nu = 0.982(17), \quad (12)$$

with $\chi_{\text{red}}^2 = 0.41$, $p = 0.75$, and compatibility with $\nu = 1$ at the level of 1.09σ (see Fig. 13).

4.3 Magnetic Exponents

The reduced susceptibility χ'_{max} and the magnetization $m(\beta_{pc})$ asymptotically scale as $L^{\gamma/\nu}$ and $L^{-\beta/\nu}$ at criticality [1, 3]. The maximum of χ' was extracted by means of a local parabolic interpolation over the three points around the sampled maximum.

The magnetization m was evaluated through spline interpolation at β_{pc} for each L , with uncertainty estimated by bootstrap. The results are reported in Tab. 3.

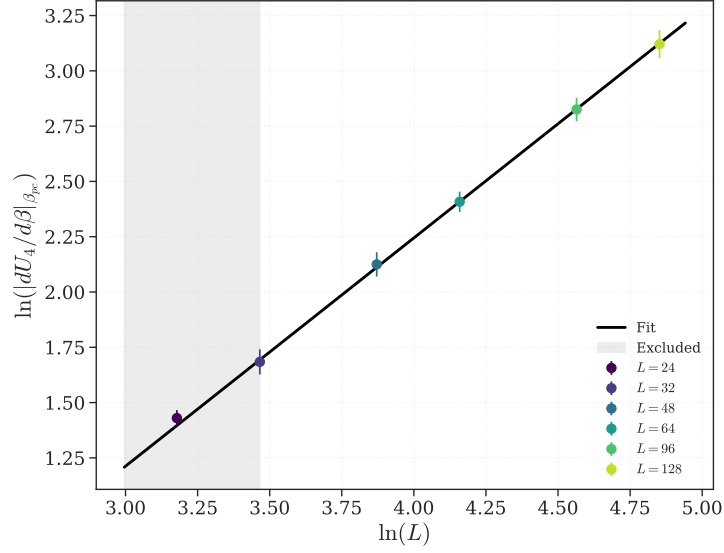


Figura 13: Log-log representation of the correlated FSS fit of the Binder slope $S(L)$, performed for $L \geq 32$.

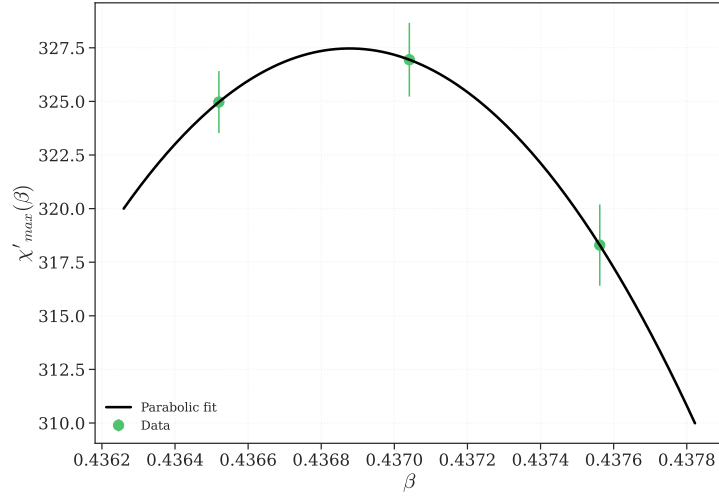


Figura 14: Extraction of the maximum $\chi'_{\max}$ by means of a local three-point parabolic fit for $L = 96$.

L	$\beta_{\text{peak}}^{(\chi')}$	$\chi'_{\max}$	$m(\beta_{pc})$
24	0.42418(86)	29.4(2)	0.67779(79)
32	0.42723(32)	48.5(2)	0.65487(84)
48	0.43225(26)	97.8(5)	0.62298(97)
64	0.43493(18)	162.2(9)	0.60114(81)
96	0.43688(08)	327.5(2)	0.57220(53)
128	0.43768(12)	543.0(3)	0.55145(69)

Tabella 3: Maxima of the reduced susceptibility χ' and values of $m(\beta_{pc}, L)$ obtained by spline interpolation.

The *leading-only* fit gives:

$$\frac{\gamma}{\nu} = 1.7425(43), \quad \chi_{\text{red}}^2 = 0.326, \quad p = 0.807, \quad (13)$$

$$\frac{\beta}{\nu} = 0.1235(11), \quad \chi_{\text{red}}^2 = 0.280, \quad p = 0.840. \quad (14)$$

The deviations from the exact values $7/4$ and $1/8$ are 1.74σ and 1.28σ , respectively. The fit is shown in Fig. 15.

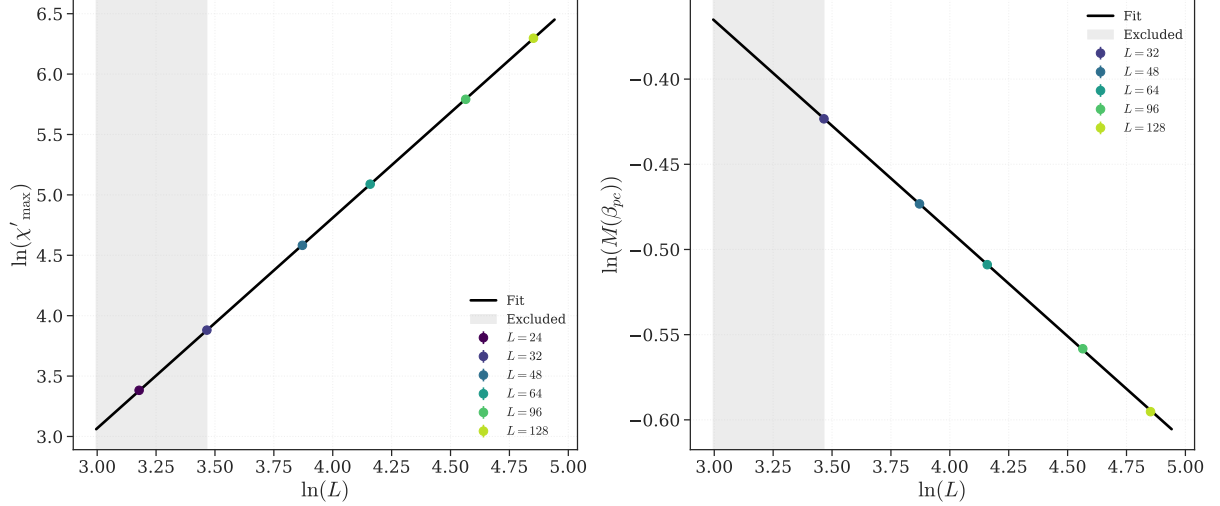


Figure 15: Log-log representation of the *leading-only* FSS fits of χ'_{max} (left) and $M(\beta_{pc})$ (right), performed for $L \geq 32$.

4.4 Analysis of Finite-Size Corrections

An F-test investigation was carried out between the models $AL^{x/\nu}$ and $AL^{x/\nu}(1 + BL^{-2})$ for all fits (see Tab. 4). The exponent $\omega = 2$ was taken as known from theory, since multi-parameter fits are difficult to handle.

Observable	<i>leading-only</i> exponent	χ_{red}^2	p_F
$S(L)$	$1/\nu = 1.0185(72)$	0.406	0.0759
$\chi'_{\text{max}}(L)$	$\gamma/\nu = 1.7425(43)$	0.326	0.489
$M(\beta_{pc}, L)$	$\beta/\nu = 0.1235(11)$	0.280	0.592

Tabella 4: Results of the *leading-only* fits, $\mathcal{O}(L) = AL^{x/\nu}$, and p -values from the F-test with respect to the correction models $\mathcal{O}(L) = AL^{x/\nu}(1 + BL^{-2})$, for $L \geq 32$.

The hypothesis $B = 0$ ($p_F > 0.05$) shows that the correction term is not statistically significant. The *leading-only* models were therefore adopted.

The choice $L_{\text{min}} = 32$ is examined in Fig. 16: including $L = 24$ leads to larger deviations.

4.5 Specific Heat and Exponent α

The maximum $C_{\text{max}}(L)$ was obtained by local parabolic fitting, analogously to the susceptibility (see Fig. 17). In the 2D Ising model, a logarithmic divergence is expected, corresponding to $\alpha = 0$ [2, 3].

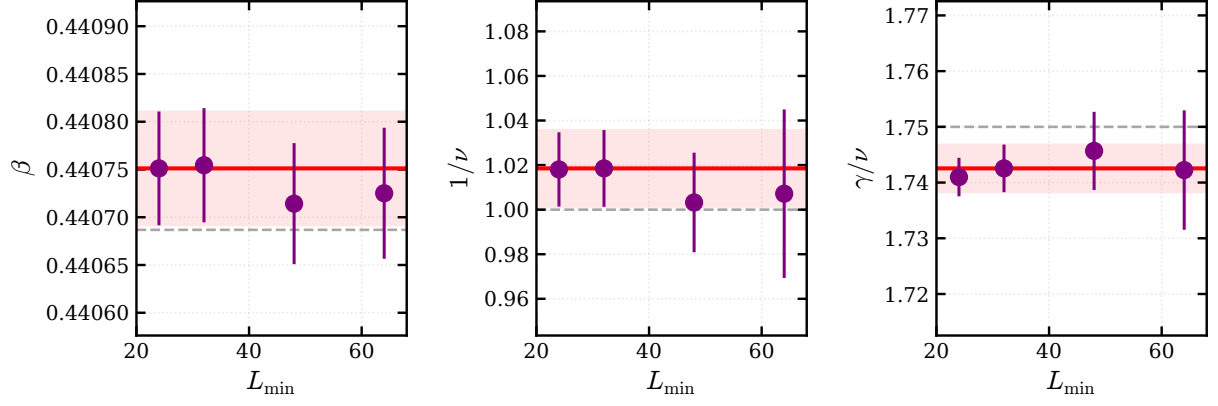


Figure 16: Stability study of the extracted exponents as the minimum cutoff $L_{\min}$ is varied.

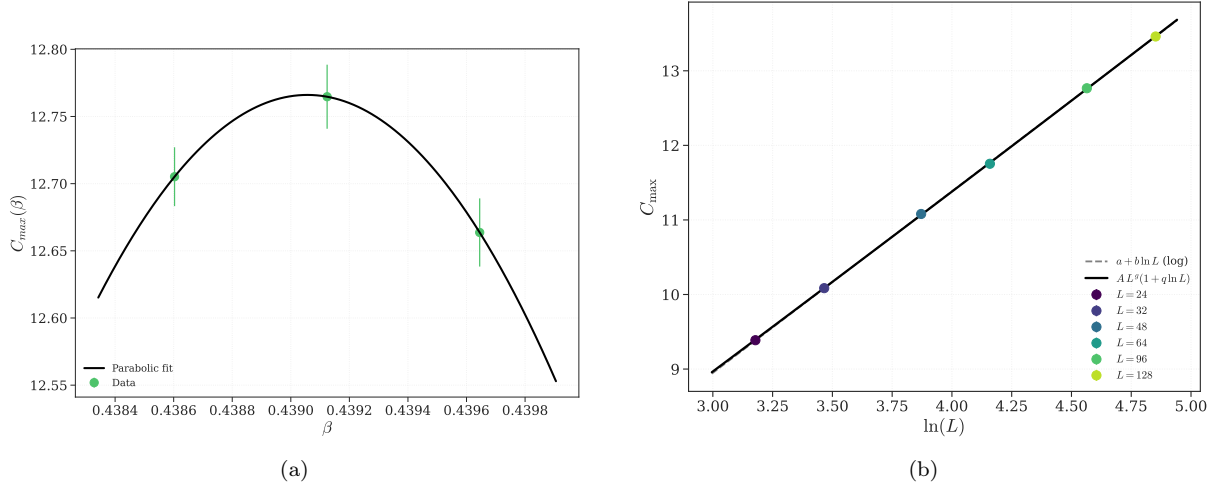


Figure 17: ((a) Extraction of $C_{\max}$ by local parabolic interpolation for $L = 96$. (b) Fit of the dependence of $C_{\max}$ on the lattice size.

The pure logarithmic fit, $C_{\max} = a + b \ln L$, yielded $\chi_{\text{red}}^2 = 0.262$ and $p = 0.853$, showing good compatibility with the logarithmic divergence. To verify this behavior without imposing $\alpha = 0$, the generalized model $C_{\max}(L) = A L^{\alpha/\nu}(1 + q \ln L)$ was considered; the result was:

$$\frac{\alpha}{\nu} = 0.0075(232), \quad \alpha = 0.0073(227), \quad \chi_{\text{red}}^2 = 0.339, \quad p = 0.712. \quad (15)$$

The result is compatible with $\alpha = 0$ within 0.32σ . The comparison between the two models gives $p_F = 0.629$, so the introduction of the free exponent α/ν does not yield a statistically significant improvement over the pure logarithmic fit (see Fig. 17b).

4.6 Statistical Correlations and Derived Exponents

The primary exponents ν , γ/ν and β/ν are obtained from the same MCMC series and were not assumed to be statistically independent. The final standard deviations are:

$$\gamma = \nu \left(\frac{\gamma}{\nu} \right) = 1.711(29), \quad (16)$$

$$\beta = \nu \left(\frac{\beta}{\nu} \right) = 0.121(02), \quad (17)$$

$$\eta = 2 - \frac{\gamma}{\nu} = 0.258(04), \quad (18)$$

$$\delta = 1 + \frac{\gamma/\nu}{\beta/\nu} = 15.10(13). \quad (19)$$

4.7 Data Collapse

The curves $\mathcal{O}(\beta, L)$, rescaled according to FSS, are expected to collapse asymptotically onto a single *master curve* [1, 3], defined in terms of the invariant variable $x = (\beta - \beta_{pc}) L^{1/\nu}$. The collapse of the magnetic observables is shown in Fig. 18, while that of the Binder cumulant and the specific heat is shown in Fig. 19.

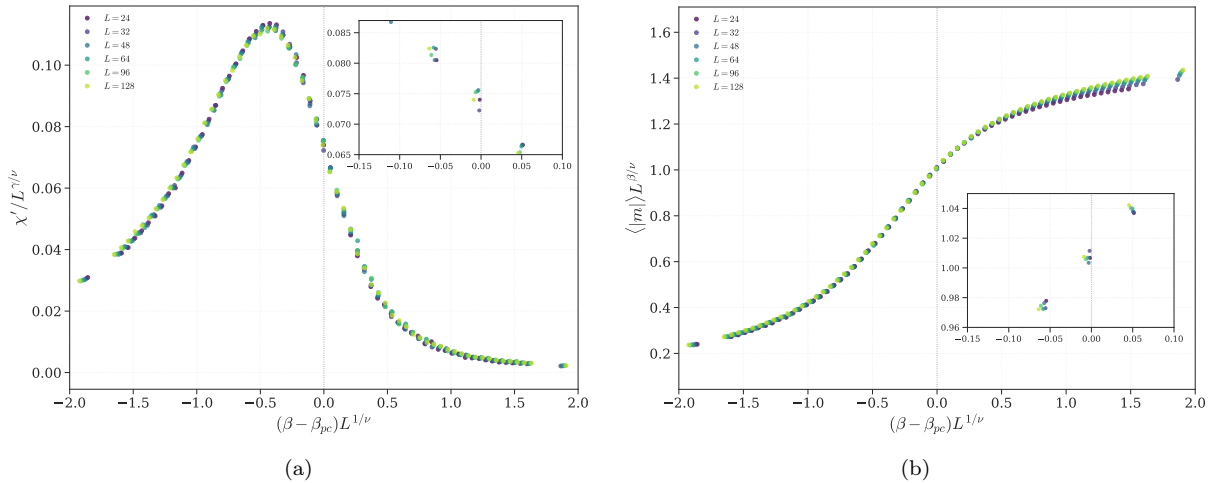


Figure 18: Data collapse for $\chi' L^{-\gamma/\nu}$ (a) and $m L^{\beta/\nu}$ (b) as a function of the scaling variable.

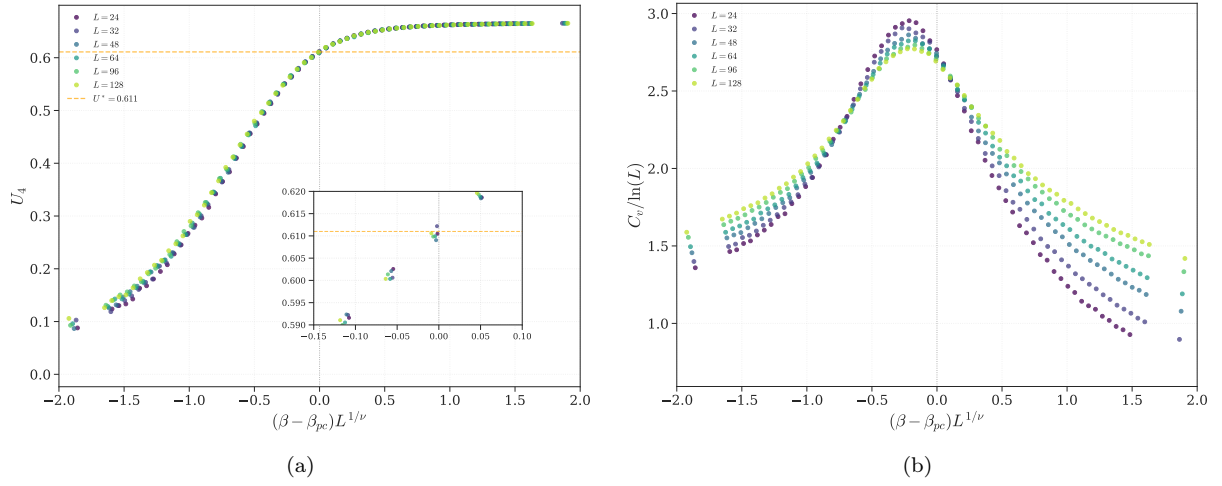


Figure 19: Data collapse of the Binder cumulant U_4 (a) and of the rescaled specific heat $C/\ln L$ (b).

4.8 Summary of Results

Tab. 5 summarizes the results of the analysis. For ν , γ/ν and β/ν , *leading-only* fits were adopted, while statistical correlations among the primary exponents were included in the propagation of derived quantities.

Quantity	Measured value	Exact/ref.
β_{pc}	0.440751(60)	0.440687
U^*	0.6116(6)	0.6107
$1/\nu$	1.019(17)	1
ν	0.982(17)	1
γ/ν	1.743(04)	1.750
β/ν	0.124(01)	0.125
α	0.007(23)	0
γ	1.711(29)	1.750
β	0.121(02)	0.125
η	0.258(04)	0.250
δ	15.10(13)	15

Tabella 5: Summary of the FSS results. The primary-exponent fits use lattice sizes $L \geq L_{\min} = 32$.

A Source code

The complete source code for the simulation and analysis is available at the following public repository:

https://github.com/Phoeredor/Ising2D_Metropolis

The repository includes: a **C** implementation for the simulations and **Python** scripts for the FSS analysis.

Riferimenti bibliografici

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